Write a 400–600-word reflective essay on ***what technical skills you and your group developed to create your AI agent?***

Your reflection should be structured using the ***"What? So What? Now What?" model*** and include specific examples from ISA104 course.

Sample Essay

What?

In ISA104, I developed technical skills for designing an AI agent through two practical activities: the Week 5 tutorial on retrieval-augmented generation (RAG) and Week 4 Lab 2: Practices on Prompt Engineering. Together, these activities showed me that an agent's performance depends not only on the language model, but also on the information it receives and the instructions it follows. During the Week 5 tutorial, I explored how RAG systems use external resources to provide a model with relevant information. I learned about semantic chunking, embeddings and vector databases. A RAG system divides documents into sections, represents those sections as embeddings, and retrieves relevant information in response to a user's query. At first, I found it difficult to understand how the size and meaning of each chunk could affect retrieval. I learned that short chunks may lose important context, whereas long chunks may include irrelevant information. In Lab 2, my group used Python scripts and the Ollama API in VS Code to set up LLM environments, revise prompts, compare model responses and test prompt injections.

So What?

Studying RAG and prompt engineering shifted my focus beyond the language model itself to the wider system that shapes an AI agent's performance. I now recognize that the quality of the retrieved information and the clarity of the instructions are also essential to an agent's reliability. RAG can provide useful information, and prompts can guide how a model responds, but neither technique guarantees accuracy. If a system retrieves irrelevant information or separates details from their context, the model may produce an unreliable answer. Similarly, a carefully written prompt does not remove the need to test how an agent responds to malicious instructions. The group work also helped me become more systematic. Specifically, when we compared responses, I learned to look beyond which answer sounded most convincing and consider what evidence supported it. I began to view testing as a way to identify weaknesses and improve an agent, rather than simply as a final check. This experience showed me that developing AI systems requires technical knowledge, critical judgement and collaboration.

Now What?

In future projects, I will begin by clarifying the agent's purpose, identifying its users, and deciding what information it needs. These decisions will guide my choices about the model, retrieval process, and prompt design. For a RAG system, I will assess source reliability and relevance, then test different chunking approaches with varied questions. I will check whether retrieved context supports answers and investigate whether errors come from the sources, retrieval, or prompts. This approach will stop me from blaming the model before examining the whole system. I will also run prompt-injection tests and compare models early in development. More importantly, as part of a team, I will record prompt changes, test results, and design decisions so that others can review and reproduce our work. Documenting this process will help me make evidence-based decisions and learn from unsuccessful tests. These activities have taught me that reliability requires ongoing evaluation, not just a final check. Therefore, I will treat testing and iteration as core design activities and use each result as feedback to improve the agent's robustness.